

API Documentation

AI-Powered Internship Performance Prediction & Risk Analytics Platform

Ezitech Engineering Framework — Case Study AI-005 · Repo: [intern-analytics-platform](#)

Submitted By: Saira Ejaz & Sadaf Riaz

1. Overview & Conventions

Base URL: `http://localhost:8000` **Interactive docs:** `/docs` (Swagger), `/redoc` **Health check:** `GET /health`

All request and response bodies are JSON. This document covers only the routers mounted in `app/main.py`: `auth`, `admin`, `interns`, `predict`, `recommendations`, `mentor`, and `students`. A file `app/routers/student.py` (singular) also exists in the repository but is never imported by `main.py` and is dead code — it is excluded here.

Authentication

Every session uses a JWT bearer token (HS256 algorithm, 8-hour expiry). A token is obtained via `POST /auth/login` or `POST /auth/register`, and must be sent on every protected request as:

Authorization: Bearer <access_token>

Roles are Admin (A), Mentor (M), and Student (S). An asterisk (*) marks an endpoint that is additionally scoped to the caller's own roster or record — a mentor or student cannot read another person's data even with a valid token for their role; the ownership check runs server-side on every request, not just in the UI. "pub" means no token is required.

Note on registration: `POST /auth/register` is public but always creates a student account it inserts both a new Intern record (unassigned, with no mentor yet) and the matching User login in a single transaction, then returns a token so the new user lands straight on their dashboard.

Error format & status codes

Errors are returned as: `{ "detail": "<message>" }`. Pydantic validation failures instead return FastAPI's standard 422 field-error array.

Code	Meaning
200	Success
401	Missing, invalid, or expired token or wrong login credentials
403	Valid token but wrong role, disabled account, or not the record owner
404	Referenced record (intern, report, submission) not found
409	Conflict duplicate email on register, or duplicate weekly report for the same week
422	Validation error malformed body, invalid enum value, or a failed business rule (e.g. password under 6 characters)

2. Endpoint Reference

Columns: Method · Path · Auth · Request body · Success response · Notes (non-200 codes and behaviour worth knowing).

2.1 Auth — /auth

Method	Path	Auth	Request body	Response	Notes
POST	/auth/login	pub	email, password	access_token, role, name, linked_id	401 / 403 (disabled account)
POST	/auth/register	pub	name, email, password, technology, batch	access_token, role, name, linked_id	409 / 422 — also creates an Intern record (see note above)
POST	/auth/change-password	any	current_password, new_password	{ detail }	401 / 422

2.2 Interns — /interns

Method	Path	Auth	Request body	Response	Notes
POST	/interns/	A	InternCreate	InternOut	—
GET	/interns/	A, M	skip, limit (query)	InternOut[]	—
GET	/interns/unassigned/list	A	—	InternOut[]	Interns where mentor_id = null (self-registered, awaiting assignment)
GET	/interns/{id}	A, M*, S*	—	InternOut	403 / 404
GET	/interns/{id}/detail	A, M*, S*	—	profile + predictions + recommendations (+ reports/projects for A/M)	403 / 404
PATCH	/interns/{id}	A	mentor_id, batch, status, expected_end_date	InternOut	Partial update, admin-only

2.3 Predictions — /predict

Method	Path	Auth	Request body	Response	Notes
POST	/predict/dropout-risk	A, M*, S*	intern_id	PredictionOut	403 / 404
POST	/predict/performance-trend	A, M*, S*	intern_id	PredictionOut	predicted_value: 0 / 1 / 2 = declining / stable / improving. 403 / 404
POST	/predict/success-probability	A, M*, S*	intern_id	PredictionOut	403 / 404
POST	/predict/learning-growth	A, M*, S*	intern_id	PredictionOut	Also saves a separate skill_growth prediction row. 403 / 404
POST	/predict/completion-probability	A, M*, S*	intern_id	PredictionOut	Derived: 1 – dropout_risk. No separate model.

POST	/predict/project-success-probability	A, M*, S*	intern_id	PredictionOut	Reuses the Success Probability model as a documented approximation.
GET	/predict/mentor-workload	A, M	—	Array of { mentor_id, name, active_interns, max_capacity, utilization_pct, overloaded }	Pure SQL — active-interns count vs. max_capacity per mentor, no ML model
GET	/predict/history/{id}	A, M*, S*	—	PredictionOut[]	403 / 404

PredictionOut = { intern_id, prediction_type, predicted_value, confidence, explanation, created_at }. Every POST endpoint above saves a new prediction row before returning it, so /predict/history/{id} always reflects the full call history for that intern.

2.4 Recommendations — /recommendations

Method	Path	Auth	Request body	Response	Notes
POST	/recommendations/{id}/generate	A, M*	—	RecommendationOut[]	403 / 404
GET	/recommendations/{id}	A, M*, S*	—	RecommendationOut[]	403 / 404

2.5 Admin dashboard — /admin

Method	Path	Auth	Response	Notes
GET	/admin/overview	A	totals, avg risk/success, dropout & success distributions, high_risk[], top_performers[], by_technology[], by_batch[]	Program-wide; high_risk = dropout_risk > 0.5, top_performers = success_probability > 0.9

2.6 Mentor dashboard — /mentor

Method	Path	Auth	Request body	Response	Notes
GET	/mentor/overview	M	—	mentor stats, roster[], weak/strong students[], recs[]	Own roster only
GET	/mentor/weekly-reports	M	—	WeeklyReport[] (last 50)	Own roster only
GET	/mentor/projects	M	—	ProjectSubmission[] (last 50)	Own roster only
GET	/mentor/reviews	M	—	unreviewed reports[] + submissions[]	Own roster only
PATCH	/mentor/reviews/weekly-report/{id}	M*	comment	{ status, report_id }	404 if not owned
PATCH	/mentor/reviews/project-submission/{id}	M*	comment	{ status, submission_id }	404 if not owned
GET	/mentor/risk-alerts	M	—	alerts[] (critical / high / moderate)	Own active interns only; critical > 0.7, high > 0.5, moderate > 0.3 dropout risk, else omitted

2.7 Student self-service — /students/me

Method	Path	Auth	Request body	Response	Notes
GET	/students/me/profile	S	—	InternProfileOut	—
PATCH	/students/me/profile	S	phone, education, skills, bio, linkedin_url, github_url	InternProfileOut	Only these fields are editable
GET	/students/me/tasks	S	—	TaskOut[]	—
PATCH	/students/me/tasks/{id}	S	status	TaskOut	404 / 422
POST	/students/me/attendance/checkin	S	—	AttendanceOut	Idempotent per day
GET	/students/me/attendance	S	—	AttendanceOut[] (last 60)	—
GET	/students/me/weekly-reports	S	—	WeeklyReportOut[]	—
POST	/students/me/weekly-reports	S	week_start_date, hours_worked, summary, challenges	WeeklyReportOut	409 if that week already has a report
GET	/students/me/projects	S	—	ProjectSubmissionOut[]	—
POST	/students/me/projects	S	title, description, repo_url, demo_url	ProjectSubmissionOut	—
GET	/students/me/overview	S	—	profile, mentor, predictions, tasks, attendance, github, comms, recs	Full dashboard aggregate

2.8 System

Method	Path	Auth	Response
GET	/health	pub	{ "status": "healthy" }
GET	/metrics	pub	Prometheus metrics (via prometheus-fastapi-instrumentator)

3. Schema Quick Reference

Pydantic models from app/schemas/schemas.py. A trailing “?” marks an optional field.

Model	Fields
InternOut	intern_id, name, email, technology, mentor_id?, batch?, start_date, expected_end_date?, status
InternProfileOut	intern_id, name, email, technology, batch?, status, start_date, expected_end_date?, phone?, education?, skills?, bio?, linkedin_url?, github_url?, avatar_color?
InternProfileUpdate	phone?, education?, skills?, bio?, linkedin_url?, github_url? (all optional)
InternUpdate	mentor_id?, batch?, status?, expected_end_date? (all optional, admin-only)
PredictionOut	intern_id, prediction_type, predicted_value, confidence?, explanation? (object), created_at
RecommendationOut	intern_id, recommendation_type, message, created_at
TaskOut	task_id, intern_id, task_name, assigned_date, due_date?, completed_date?, status, difficulty?
AttendanceOut	attendance_id, intern_id, date, present (bool)
WeeklyReportOut	report_id, intern_id, week_start_date, hours_worked?, summary, challenges?, created_at
ProjectSubmissionOut	submission_id, intern_id, title, description?, repo_url, demo_url?, submitted_at

Role → endpoint group matrix

Router	Admin	Mentor	Student
/auth (login, register)	public	public	public
/interns	full	own roster (read)	own record (read)
/predict	full	own roster	own record
/recommendations	full	own roster	own record (read)
/admin	yes	no	no
/mentor	no	yes (own data)	no
/students/me	no	no	yes (own data)